

Fernando Hidalgo Pineda

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EDUCATION

PhD at Astronomisches Rechen-Institut (ARI), University of Heidelberg <i>Advisor:</i> Dr. Max Grönke	Heidelberg, Germany September 2024 - Present
External Master's Project: "Bridging scales for galactic winds in cosmological simulations" <i>Advisors:</i> Prof. Romeel Davé, Dr. Britton Smith	Edinburgh, UK August 2023 - June 2024
University of Glasgow MSci Physics with Astrophysics, First-Class Honours	Glasgow, UK September 2019 - June 2024

RESEARCH EXPERIENCE

Institute of Astronomy, University of Cambridge <i>Advisors:</i> Prof. Debora Sijacki, Dr. Martin Bourne	Cambridge, UK June 2023 - Present
Max Planck Institute for Astrophysics <i>Advisors:</i> Dr. Ryan Farber, Dr. Max Grönke	Garching, Germany June 2022 - April 2023
Institute of Astrophysics of Andalusia, IAA-CSIC <i>Advisors:</i> Prof. Concepción Del Olmo	Granada, Spain Sep 2017 - Mar 2018
UAL School of Physics and Chemistry, Calar Alto Observatory <i>Advisor:</i> Dr. María José García Salinas	Almería, Spain Jun 2017 - Sep 2017
Faculty of Physics, University of Granada <i>Advisor:</i> Prof. Alonso Salinas Extremera	Granada, Spain Sep 2016 - Mar 2017

PUBLICATIONS

Hidalgo-Pineda, F., Bourne M., Sijacki D., *Impact of a Bipolar AGN-driven Wind Model upon Galactic Evolutionary Features*, In prep., submission in 2026 to MNRAS.

Hidalgo-Pineda, F., Gronke M., Grete P., *The Launching of Galactic Winds from a Multiphase ISM*, 2025, sub. to MNRAS, arXiv preprint: [10.1093/mnras/stad3069](https://arxiv.org/abs/10.1093/mnras/stad3069).

Hidalgo-Pineda, F., Farber R. J., Gronke M., *Better Together: The Complex Interplay Between Radiative Cooling and Magnetic Draping*, 2023, MNRAS, 527, 135, doi: [10.1093/mnras/stad3069](https://arxiv.org/abs/10.1093/mnras/stad3069).

INVITED & CONTRIBUTED TALKS

Nov 2025 , "From a Spherical Cow to a Herd of Real Ones"	MMAM Conference, Tegernsee, Germany.
Oct 2025 , "The Formation of Outflows in a Fractal ISM"	Contributing Speaker, MMC Conference, Olbia, Italy.
Sept 2025 , "From Spherical Cows to a Herd of Real Ones"	Contributing Speaker, MIST Conference, Cargese, France.
May 2025 , "Multiphase Outflows in Realistic Environments"	Guest Speaker, ARI/ZAH Heidelberg, Germany.
Dec 2024 , "Cold Outflows and Their Interaction with Magnetic Fields"	Invited Speaker, ARI/ZAH Heidelberg, Germany.
July 2023 , "Can AGN-Driven Winds Change Our Galactic Evolution Models?"	IoA Summer Seminar Series, Cambridge, UK.
Feb 2023 , "Entrainment of Cold Gas by Hot Plasmas"	Guest Talk, Edinburgh, UK.
Jan 2023 , "The Interplay Between Magnetic Draping and Radiative Cooling"	241 st AAS Meeting, Seattle, USA.
Mar 2021 , "Optimising Logistic Strategies from Short-Scale Variations in Stochastic System"	European Cassini Competition, Brussels, Belgium.
May 2018 , "How Do Galaxies Evolve? A Study of Galactic Sequences"	CSIC Project Conference, Granada, Spain.

AWARDS & GRANTS

IMPRS Fellowship	2024
IoA Undergraduate Intership Funding Programme	2023
Academic Excellence Award	2023
Cassini Hackathon National Prize, EU	2021
IV Concurso de Buenas Practicas, Spanish Ministry of Sciences	2020
Full Advanced Studies Scholarship, UK	2019

Academic Honours Award, Ministry of Education	2019
'Hands on Particle Physics' Grant	2018

PI CONFERENCES AND MEETINGS

Numerical and Theory Journal Club - ARI, Heidelberg	January 2026 - Present
Conference Modelling Multiphase Astrophysical Media - Tegernsee, Germany	November 2025
Galaxy Evolution Coffee Seminar Series - ESO, Munich	January - June 2024

CO-CURRICULAR ACTIVITIES

Sports:	Player of Kirchheim Basketball in the Regionalliga, alumni of Glasgow Lions Basketball team competing in the Scottish National League, Welfare Convenor of the Men's Basketball University Club.
Music:	Violinist with Certified Professional Music Studies at the Angel Barrios Conservatory (Spain), supplementary studies in Piano and Ukulele.
Academic:	Vice-president of Glasgow University for Sustainable Development Goals, Partner Liaison of Spanish Society, elected Class Representative.
Languages:	Spanish (Native), English (Fluent), French (Proficient), Italian (Competent).
Volunteering:	Volunteer of Amnesty International.